Supplementary material: Predict-then-Optimize Framework for Public Transport Line Redesign under Fluctuating Traffic Conditions

ABSTRACT

This document provides supplementary derivations, algorithmic and evaluation details, and additional results for the conference paper.

A. Additional related work

Previous work related to ours typically focuses on either *network design* at strategic and tactical levels, where PT lines are planned from scratch, or *reactive operational control*, which manages daily operations without altering the network structure. *Network redesign* occupies an intermediate position by adapting existing networks rather than starting anew. Within this domain, *offline redesign* relies on re-optimization for known future scenarios, whereas emerging *proactive redesign* approaches anticipate short-term uncertainty. However, these proactive methods often overlook the critical trade-off between responsiveness and structural stability, a gap that this paper specifically addresses.

A.1. Public Transport Network Design under Nominal Conditions

The PT Network Design Problem (TNDP) consists of designing line routes and their frequencies, balancing user- and operator-centric objectives, and has been studied since the foundational formulations of Ceder and Wilson (1986) through successive reviews (Guihaire and Hao, 2008; Kepaptsoglou and Karlaftis, 2009; Durán-Micco and Vansteenwegen, 2022) and reference texts (Desaulniers and Hickman, 2007; Ceder, 2016),Gkiotsalitis (2020).

Most of these approaches formulate public transport networks *from scratch*, without reference to any pre-existing service configuration; although some formulations begin from a predefined set of candidate lines (Cervantes-Sanmiguel, Chavez-Hernandez and Ibarra-Rojas, 2023), the network topology is typically treated as a fresh design decision rather than an adaptation of an existing operational configuration. These methods are primarily intended for long-term strategic or tactical planning. The TNDP is NP-hard (Magnanti and Wong, 1984).

User-centric performance has been expressed in numerous ways. In addition to overall travel duration (John, Mumford and Lewis, 2014; Heyken Soares, Mumford, Amponsah and Mao, 2019), recent studies also examine network coverage (Cadarso, Codina, Escudero and Marín, 2017), route efficiency (Suman and Bolia, 2019), and demand fulfillment (Barahimi, Eydi and Aghaie, 2021). Operator-centric objectives are focused on limiting operational and capital expenditures (Ahern, Paz and Corry, 2022; Chen, Liu, Wang and Chen, 2018; Calabrò, Araldo, Oh, Seshadri, Inturri and Ben-Akiva, 2023), or proxies for them, such as the overall length of line routes. Balancing these objectives draws on two distinct methodological strands. *Multi-level* formulations capture hierarchical decision-making among different agents: Szeto and Jiang (2014) established a bilevel framework in which an upper-level planner minimizes passenger transfers and a lower-level model handles PT assignment; Tian, Wang and Lin (2021) extended this to a trilevel structure with passenger routing strategies at an intermediate level. *Multi-objective* formulations (as ours) instead seek Pareto-optimal trade-off solutions across conflicting criteria: Cervantes-Sanmiguel et al. (2023) analyzed the trade-off between travel-time reduction and fare reduction.

Note that all these methods design PT network based on *nominal* (e.g., average, peak, etc.) *traffic conditions*. Therefore, as noted by (Tan, Peng, Huang, Wang, Qiu and Liu, 2024), the resulting networks may be inefficient faced with real-time variations in traffic patterns, deviating from nominal conditions.

A.2. Reactive Operational Control under Disruptions and Uncertainty

In most real-world operations, the structure of the public transport network remains fixed as originally planned. When deviations from nominal conditions occur due to exogenous factors (Nissen, Becker, Dähne, Rabe, Scheffler,

Solle and Ulbrich, 2020; Chen, Yu, Zhang and Guo, 2009), operators rely on short-term corrective actions to mitigate supply–demand mismatches. Typical examples include stop-skipping (Peled, Lee, Jiang, Dauwels and Pereira, 2021), short-turning, speed changing, holding (Tang, Ceder and Ge, 2018; Manasra and Toledo, 2019; Gkiotsalitis, 2020; Olvera-Toscano, Ríos-Solís, Ríos-Mercado and Sánchez Nigenda, 2023), detouring of interrupted lines (Zheng, Sun, Kang, Dai and Wu, 2023), and replacement buses. Each control action has distinct disadvantages. Namely, bus holding increases the inconvenience of on-board passengers at the holding stops, short-turning requires passengers to disembark and wait for a subsequent bus trip, rescheduling affects the timetables and the crew/vehicle schedules; speed control cannot be implemented when the required speeds are beyond the safety limits, and stop-skipping results in refused passenger boardings (Gkiotsalitis, 2021). While some works integrate real-time data into operational decision-making, such as schedule adjustment for enhanced reliability (Cao, Ceder and Zhang, 2019), these frameworks focus on timetable or speed control within a fixed route structure and do not explicitly consider structural reconfiguration of the network topology under time-varying conditions. It should be noted that proactive network redesign, as proposed in this work, may generate analogous and in some respects stronger inconveniences on the operational side, affecting vehicle scheduling and driver rostering, in particular with human-driven PT vehicles; it is thus important to limit excessive structural instability (as discussed in the conference paper).

More importantly, when actual conditions deviate significantly from the nominal ones, such corrective actions may not be sufficient to restore performance (Guo, Araldo and Yacoubi, 2024), since the overall structure is maintained as planned. Strategies such as stop-skipping and short-turning have also appeared in the context of offline or preplanned redesign, to generate alternative route configurations or long-term service plans under hypothetical disruption scenarios. They are outside the scope of this paper, which instead focuses on proactive redesigning lines for the hour ahead based on short-term forecasts.

A.3. Offline and Preplanned Network Redesign under Known Scenarios

Various approaches can be employed in the bus route adjustment problem to generate adaptive bus routes, including rerouting, short-turning, skip-stopping, and branching (Cao and Ceder, 2019; Ma, Yang and Li, 2023; Wu, Kulcsár, Qu et al., 2021). In contrast to the network design approaches described in §A.1, these studies explicitly start from an *existing* public transport network and allow service or structural modifications under a limited set of anticipated scenarios. Previous research on bus route localized adjustments has often aimed at two main objectives: accommodating demand fluctuations and enhancing integration with subway networks.

Guan, Xiang, Wang, Yan and Zhao (2023) introduced a bilevel planning model to optimize customized bus routes offline based on a spatiotemporal state network. While effective for static planning, this method is impractical for real-time dispatching. Concretely, it relies on a computationally expensive nested algorithm and requires a full re-solve for any new event. Zhang, Jia, Bai, Mao, Ma and Zhang (2018) investigated long-distance replacement bus services to improve coordination with urban rail networks. While effective under static demand assumptions, this approach is inherently offline and cannot be applied in real-time operational contexts. When passenger flow control is implemented in a subway system, Zhou, Hu, Huang and Deng (2021) optimized existing bus routes to transport passengers temporarily restricted from entry, in case of disruptions. Wang, Chen, Yu, An and Wu (2022) proposed a two-stage method to mitigate disruptions to bus routes under planned construction scenarios, relying on predefined work-zone impacts and static travel-time estimates. Wu, Zuo, Zhou, Wan, Zhao and Yang (2024) proposed a multi-objective ant colony algorithm to adjust existing bus routes and introduced a sequence-level similarity to prevent the redesigned network from being excessively different from the original. However, their approach remains static and cannot capture time-varying adaptations. Zheng, Sun, Wu and Kang (2024) proposed localized adaptive bus routes under anticipated disruptions, while Zhang, Xu and Wang (2024) addressed large-scale emergency disruptions assuming available spare capacity. These assumptions do not hold under recurrent traffic fluctuations, where no redundant lines exist and adaptation must occur within the same constrained topology.

A.4. Network Redesign under Predictive Uncertainty and Structural Consistency

Traditional network design handles uncertainty primarily by optimizing for expected scenarios (stochastic optimization) or worst-case bounds (robust optimization) (Bertsimas, Gupta and Kallus, 2018; Wang and Qi, 2020); see Ge, Voß and Xie (2022) for a recent review of robustness and disturbances in public transport. However, these approaches typically determine a single network configuration before operations begin, with limited adaptation to realized road conditions (i.e., without reconfiguration). While this ensures stability and ease of operations, it may become ineffective in preserving performance faced with high traffic fluctuations. Another line of work studies topological metrics to

evaluate network resilience (Massobrio and Cats, 2024; Pu, Li and Ma, 2022), e.g., they measure vulnerability after a perturbation occurs. These approaches remain purely diagnostic and descriptive, rather than being prescriptive (as the approach we present in this paper), in that they do not provide guidance on the optimization of the network design.

A.5. Vehicle Routing Problems and Demand-Responsive Transport (DRT)

In Vehicle Routing Problems (VRPs) and DRT, vehicle routes are constructed on-demand and solution stability has been employed to keep revised plans close to a baseline (Touzout, Ladier and Hadj-Hamou, 2021). However, this stability applies to routing or scheduling assignments rather than the structure of the PT line network itself. Moreover, DRT and other VRP-based systems do not exhibit a structured network of lines, and therefore fail to achieve sufficient passenger consolidation (Calabrò, Le Pira, Giuffrida, Inturri, Ignaccolo and Correia, 2022). Hybrid fixed/flexible services partly bridge this gap (Calabrò et al. (2023)), but still keep the fixed lines inflexible and flexible routes unstructured. By contrast, we study a high-capacity mass-transit system organized around a structured network of fixed lines, where existing lines are reconfigured rather than routes being computed from scratch. Our approach is therefore conceptually distinct from VRP- or DRT-based formulations. A direct comparison with DRT is also inappropriate, since we address high demand levels typical of dense urban cores, for which DRT is generally unsuitable in terms of both computational scalability and service performance (Calabrò et al., 2022).

A.6. Research gaps

Table 1 summarizes related work. Most existing studies treat PT network design as a deterministic optimization problem, ignoring the inherent uncertainty and variability in urban traffic conditions. Only a few studies consider any form of uncertainty, and few redesign the network in response to short-term forecasts while also modeling travel-time variability in a systematic way. An exception is Guo, Araldo and El Yacoubi (2025), where, however, (i) future travel times on road links were assumed to be perfectly known (which is unrealistic), so the impact of prediction errors was not considered, while we model the error of a well established prediction model, (ii) only a single simplified objective was adopted, while we solve a multiobjective optimization, and (iii) their approach is tested only on a simple hypothetical scenario, while we consider real-world dataset to fit the prediction error models.

zg~~To our knowledge, this work is the first to systematically study the potential of proactively redesigning PT lines (e.g., for the hour ahead) under fluctuating traffic.~~This work studies the potential of proactively redesigning PT lines (e.g., for the hour ahead) under fluctuating traffic. Redesign decisions are based on short-term forecasts, are triggered when needed (i.e., when traffic conditions are highly perturbed causing user- and operator-performance degradation), and balance passenger travel time against operational cost through controlled structural changes.

B. Detailed line-structure and fleet constraints

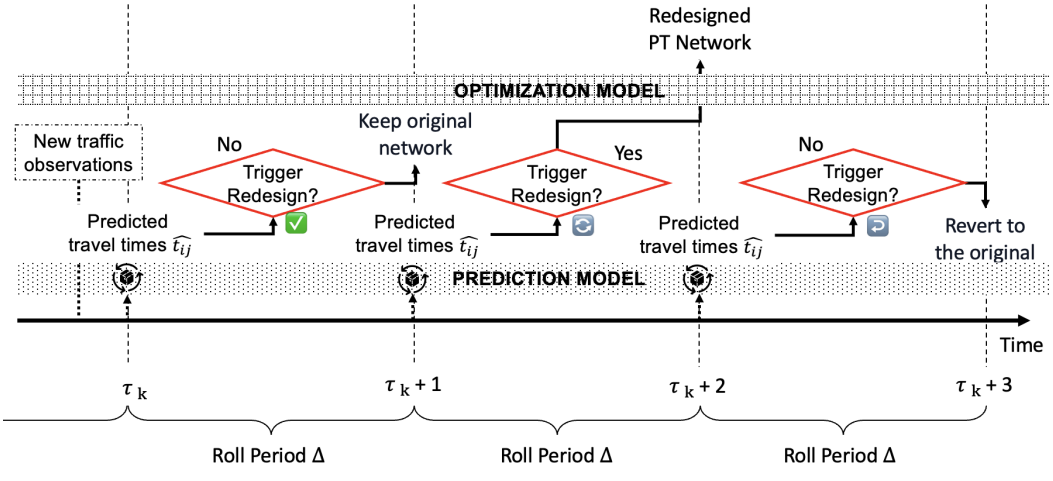


Figure 1: Proactive Line Redesign Framework.

Study	Solution Method	Online/Proactive Redesign	Similarity Control	Decision Focus
(Magnanti and Wong, 1984)	Exact methods	No	No	ND
(John et al., 2014)	Multi-objective GA	No	No	ND
(Heyken Soares et al., 2019)	Scaled network GA	No	No	ND
(Cadarso et al., 2017)	Branch-and-cut	No	No	ND
(Ahern et al., 2022)	Gradient-based	No	No	ND
(Tan et al., 2024)	Congestion analysis	No	No	FS
(Guan et al., 2023)	Custom bilevel	No	No	OR
(Zhang et al., 2018)	Tailored GA	No	No	OR
(Wang et al., 2022)	Two-stage heuristic	No	No	DR
(Wu et al., 2024)	Ant Colony	No	NW	OR
(Zheng et al., 2024)	Heuristic	No	No	DR
(Cao and Ceder, 2019)	Skip-stop heuristic	No	No	VC
(Ma et al., 2023)	Integer programming	No	No	OR
(Tang et al., 2018)	Optimal control	No	No	VC
(Manasra and Toledo, 2019)	Branch-and-bound	No	No	VC
(Gkiotsalitis, 2020)	Evolutionary	No	No	VC
(Olvera-Toscano et al., 2023)	Holding optimization	No	No	VC
(Zheng et al., 2023)	Coordination heuristic	No	No	VC
(Guo et al., 2025)	Exact methods	Partial	Penalty	NR
Our Approach	**NSGA-III based optimization**	**Yes**	**Jaccard**	**NR**

Notes: Online/Proactive Redesign – **No**: no online or proactive redesign; **Partial**: limited redesign capability (e.g., sampling-based or scenario-specific updates); **Yes**: proactive redesign enabled (e.g., predict-then-optimize). Similarity Control – **No**: no explicit similarity mechanism; **NW**: Needleman–Wunsch-based similarity; **Penalty**: penalty-based similarity; **Jaccard**: Jaccard-based similarity. Decision Focus – **ND**: from-scratch network design; **FS**: frequency setting; **OR**: offline or preplanned redesign of an existing network; **DR**: perturbation-oriented redesign or response; **VC**: vehicle or operational control; **NR**: network redesign under uncertain travel times.

Table 1
Comparative analysis of public transport optimization approaches across decision scope, online/proactive redesign capability, and similarity control mechanisms.

zgThe conference paper presents a compact version of the design constraints. The following formulation retains the more detailed line-structure constraints from the original manuscript. Let $\mathcal{N}$ be the set of stops, $\mathcal{E}$ the candidate physical links, and $\mathcal{L}$ the lines; $\vec{\mathcal{E}}$ contains both directed arcs for each physical link. Binary variables y^ℓ, a_i^ℓ, b_i^ℓ, and e_{ij}^ℓ encode line activation, terminals, and directed path arcs. The integer n^ℓ is the number of vehicles on line ℓ, $\bar{n}$ is its upper bound, and $V_{\max}$ is the total fleet budget. The order variable u_i^ℓ eliminates subtours.

$$\sum_{i\in\mathcal{N}} a_i^\ell = y^\ell, \qquad \sum_{i\in\mathcal{N}} b_i^\ell = y^\ell \qquad \forall \ell \in \mathcal{L} \tag{1}$$

$$a_i^\ell + b_i^\ell \le 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{2}$$

$$e_{ij}^\ell + e_{ji}^\ell \le y^\ell \qquad \forall \{i,j\} \in \mathcal{E},\ \forall \ell \in \mathcal{L} \tag{3}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell + a_i^\ell = \sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell + b_i^\ell \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{4}$$

$$\sum_{j:\{i,j\}\in\mathcal{E}} e_{ij}^\ell \le 1, \qquad \sum_{j:\{i,j\}\in\mathcal{E}} e_{ji}^\ell \le 1 \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{5}$$

$$u_i^\ell - u_j^\ell + 1 \le |\mathcal{N}|\left(1 - e_{ij}^\ell\right) \qquad zg\cancel{\forall \{i,j\} \in \mathcal{E}} \forall (i,j) \in \vec{\mathcal{E}},\ \forall \ell \in \mathcal{L} \tag{6}$$

$$1 \le u_i^\ell \le |\mathcal{N}| \qquad \forall i \in \mathcal{N},\ \forall \ell \in \mathcal{L} \tag{7}$$

$$n^\ell \ge y^\ell, \qquad n^\ell \le \bar{n}\, y^\ell \qquad \forall \ell \in \mathcal{L} \tag{8}$$

$$\sum_{\ell \in \mathcal{L}} n^{\ell} \leq V_{\max} \tag{9}$$

Constraints (1)–(2) synchronize line activation with terminal selection, mandating that every active line connects exactly one initial terminal to one final terminal, thereby preventing zero-length loops. Constraint (3) ensures that an inactive line has no links and an active line uses at most one direction of each physical link in its path representation. Vehicles traverse the physical line in both directions. Flow conservation (4) and degree constraints (5) restrict the line to a non-branching path. The ordering constraints (6)–(7) eliminate disconnected subtours. Resource constraints (8)–(9) assign vehicles only to active lines and enforce the global fleet budget.

zgThe ordering inequality applies to both candidate arc directions for each physical link. The physical edge indicator used in the line-labelled Jaccard objective is $x^{\ell}_{ij} = e^{\ell}_{ij} + e^{\ell}_{ji}$, so reversing the path representation alone does not change overlap. The conference paper defines the three objectives and the induced shortest-path passenger evaluation.

C. Additional details of the solution algorithm

C.1. Solution Encoding

In the proposed NSGA-III method, each PT line ℓ is represented as an ordered simple path $p_{\ell} = (i_1, i_2, \ldots, i_{n_{\ell}})$ with $i_j \in \mathcal{N}$ and no repeated nodes, which enforces acyclicity by construction as required by the line-structure and fleet constraints. PT lines may either be constructed from *scratch* or obtained by *reconstructing bus routes from the existing network*. Starting from a current node i, the algorithm maintains a visited-node set $\mathcal{V}^{\ell}_{\text{visited}}$ and defines the set of feasible candidates as $\mathcal{N}_{\text{unvisited}}(i) = \{\, j \in \mathcal{N}(i) \,:\, j \notin \mathcal{V}^{\ell}_{\text{visited}} \,\}$, which is used whenever a new path segment needs to be generated. For lines constructed from scratch, the path is iteratively extended by selecting the next node $j \in \mathcal{N}_{\text{unvisited}}(i)$ according to the travel-time weighted probability

$$\mathbb{P}(j \mid i) = \frac{1/t_{ij}}{\sum_{k \in \mathcal{N}_{\text{unvisited}}(i)} 1/t_{ik}}, \tag{10}$$

which favors temporally efficient connections while preserving stochastic exploration. When a line is derived from the existing network, local modifications are applied by replacing a selected route segment with an alternative feasible path that preserves the original endpoints.

Simultaneously, vehicle allocation is managed by assigning integer variables n^{ℓ} to each line. During initialization, each line is pre-allocated one vehicle to guarantee basic service availability. The remaining $V_{\max} - |\mathcal{L}|$ vehicles are then distributed across lines proportionally to their initial one-way travel times t^{ℓ} (computed based on nominal link travel times t_{ij}).

C.2. Evolutionary Search Procedure

Each individual encodes a complete PT network design, comprising a set of line path representations and the associated vehicle allocations. Starting from an initial population, the algorithm evolves solutions in a generational manner. At each generation, offspring are produced through selection, variation, and evaluation, followed by the NSGA-III reference-point-based *environmental selection*, which prioritizes non-dominated solutions while maintaining diversity along the Pareto front. The conference paper summarizes the overall procedure.

C.2.1. Population Initialization

During population initialization , each individual is generated using the line construction rules described in § C.1. PT lines are either constructed from scratch or obtained by locally perturbing routes reconstructed from the existing network. Given a set of lines, vehicle allocations are initialized by assigning one vehicle to each active line to guarantee basic service availability. The remaining fleet is then distributed across lines according to the chromosome encoding scheme described in § C.1, ensuring $\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$.

To expedite initialization, candidate solutions are screened via a **connectivity check** on the undirected graph induced by $\mathcal{E}_{\text{PT}} = \bigcup_{\ell \in \mathcal{L}} \mathcal{E}_{\ell}$, i.e., the union of all active line edges in $\mathcal{G}_{\text{PT}}$. By employing a Union–Find data structure, we efficiently identify connected components, and verify that, for every OD pair $k \in \mathcal{K}$, the origin o_k and destination d_k belong to the same connected component (which implies that it is possible to go from one to the other). The procedure is repeated until *PopSize* individuals (i.e., PT network structures) are generated that satisfy: (i) structural validity (lines are simple-paths, with no branches nor cycles), (ii) connectivity-based serviceability, i.e., all OD pairs

$k \in \mathcal{K}$ are connected within $\mathcal{G}_{\mathrm{PT}}$, and (iii) similarity, i.e., individual $\mathcal{G}_{\mathrm{PT}}$ is not extremely different from original PT network $\mathcal{G}_{\mathrm{PT}}^{\mathrm{org}}$, i.e., the Jaccard similarity index must be above threshold $J_{\min}$. These criteria enforce a necessary (but not sufficient) condition for OD reachability; full feasibility and network performance are subsequently evaluated using the decode-and-assign objective.

C.2.2. Crossover Operation

With probability P_c, a crossover operator is applied to two parent solutions, denoted as parents A and B. Each solution consists of a fixed set of PT lines, where each line is treated as a basic inheritance unit characterized jointly by its bus route topology and vehicle allocation. The crossover exchanges complete lines between parents as the primary recombination mechanism; any subsequent modifications are limited to feasibility restoration.

Line inheritance is governed by a binary selection mask $\boldsymbol{\xi} = (\xi_\ell)_{\ell \in \mathcal{L}}$, where each component is independently drawn as $\xi_\ell \sim \mathrm{Bernoulli}(0.5)$, so that both parents are treated symmetrically and structural diversity is promoted at the line level. Given $\boldsymbol{\xi}$, the offspring design is defined by block-wise inheritance:

$$(e_{ij}^{\ell})^{\mathrm{off}} = \xi_\ell (e_{ij}^{\ell})^A + (1-\xi_\ell)(e_{ij}^{\ell})^B, \qquad \forall \ell \in \mathcal{L},\ (i,j) \in \mathcal{E}, \tag{11}$$

$$(n^{\ell})^{\mathrm{off}} = \xi_\ell (n^{\ell})^A + (1-\xi_\ell)(n^{\ell})^B, \qquad \forall \ell \in \mathcal{L}. \tag{12}$$

Whenever a line topology is inherited from a parent, its vehicle allocation is inherited jointly, thereby preserving the coupling between bus route geometry and service frequency.

If any inherited line becomes structurally infeasible (e.g., disconnected or discontinuous), a feasibility-oriented restoration operator is applied to recover a valid simple-path representation using a feasibility-oriented repair procedure. Finally, fleet constraints ($n^{\ell} \geq 1$ for all active lines and $\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$) are enforced via minimal ± 1 adjustments on a small subset of lines. This correction is a feasibility projection and is not guided by objective improvement.

C.2.3. Mutation Operation

With probability P_m, a mutation perturbs a single line ℓ via local structural modifications, including node insertion, node deletion, or segment reconnection. These operations introduce stochastic variations in line topology while preserving the overall chromosome structure. If the mutation results in a disconnected or discontinuous line, feasibility is restored by reconnecting the affected segments using a feasibility-oriented repair procedure. This repair step is feasibility-oriented and does not explicitly consider objective values. Finally, global fleet conservation ($\sum_{\ell \in \mathcal{L}} n^{\ell} = V_{\max}$) is enforced in the same manner as described in § C.2.2.

C.2.4. Environmental Selection

After offspring generation, parent and offspring populations are merged and ranked via non-dominated sorting. Environmental selection follows the standard NSGA-III mechanism, employing a set of reference points N_{ref} to maintain population diversity (Deb and Jain, 2014).[1]

D. Proof and discussion of pairwise stability

For any two redesigns, $J(E^{(k)}, E^{(k+n)}) \geq J(E^{(k)}, E^{(0)}) + J(E^{(k+n)}, E^{(0)}) - 1$. This is a conditional bound and does not impose a minimum overlap by itself.

Proof. Setting $A = \mathcal{E}^{(k)}$, $B = \mathcal{E}^{(0)}$, and $C = \mathcal{E}^{(k+n)}$. The Jaccard distance between two sets X and Y is

$$d_J(X,Y) = 1 - \frac{|X \cap Y|}{|X \cup Y|} = 1 - J(X,Y).$$

Levandowsky and Winter (Levandowsky and Winter, 1971) noted that this dissimilarity measure satisfies the triangle inequality on finite sets, so that d_J may be regarded as a distance function. Hence,

$$d_J(A,C) \leq d_J(A,B) + d_J(B,C).$$

[1] The implementation details are provided in App. F.3.

Substituting $d_J(X,Y) = 1 - J(X,Y)$ into the above inequality yields

$$1 - J(A,C) \leq [\,1 - J(A,B)\,] + [\,1 - J(B,C)\,],$$

which is equivalent to

$$J(A,C) \geq J(A,B) + J(B,C) - 1.$$

zg~~, gives the desired result.~~which gives the desired result. ■

Remark. zg~~This implies that if two consecutive designs exhibit a similarity of ≥ 90% relative to the original (consistent with our numerical results), the similarity between these two consecutive designs is guaranteed to be at least 80%.~~For example, if two designs each have similarity of at least 90% to the original, their mutual similarity is at least 80%. This conditional example does not assert that every reported design meets the 90% threshold.

E. Stochastic Modelling of Travel Times and Prediction Errors

E.1. Parameter Derivation for Log-Normal Distribution

Log-normal parameter derivation. For a lognormal random variable $X \sim \text{LogNormal}(\mu, \sigma^2)$, the median is $\text{Med}(X) = \exp(\mu)$, so specifying the median m yields

$$\mu = \ln(m).$$

The mean and variance of X are

$$\text{E}[X] = \exp\Big(\mu + \tfrac{1}{2}\sigma^2\Big), \qquad \text{Var}(X) = \big(\exp(\sigma^2) - 1\big)\exp\big(2\mu + \sigma^2\big).$$

Hence the coefficient of variation is

$$\begin{aligned} cv &= \frac{\sqrt{\text{Var}(X)}}{\text{E}[X]} \\ &= \frac{\sqrt{\big(\exp(\sigma^2) - 1\big)\exp(2\mu + \sigma^2)}}{\exp(\mu + \tfrac{1}{2}\sigma^2)} \\ &= \sqrt{\exp(\sigma^2) - 1}. \end{aligned}$$

Equating this expression with the prescribed value of cv gives

$$cv^2 = \exp(\sigma^2) - 1 \qquad \Longrightarrow \qquad \sigma^2 = \ln\big(1 + cv^2\big) \qquad \Longrightarrow \qquad \sigma = \sqrt{\ln\big(1 + cv^2\big)}.$$

■

E.2. zgTraffic Speed Prediction and Error Estimation

To generate realistic stochastic prediction errors, we employ the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Li, Yu, Shahabi and Liu, 2017) to model spatiotemporal traffic dependencies. Two benchmark datasets are utilized: **METR-LA** (4 months of highway data from 207 sensors) as a proxy for Mandl's network, and **Q-Traffic** (61 days of Beijing 6th Ring Road segments) for the real-world case study. We split both datasets, chronologically, as recommended by Li et al. (2017) and Lu, Lv, Xie, Du, Xiong, Sun and Wang (2022), respectively: We train each prediction model on the first part of the chronologically ordered data and measure prediction errors on the last part, i.e., the test set. Table 2 reports the one-hour-ahead prediction errors used in our experiments.

The prediction models output traffic speeds. Since the downstream public-transport redesign model operates on travel times, the extracted prediction residuals must be expressed in travel-time space. If a link length d is available, the realized and predicted travel times are $T = d/v$ and $\hat{T} = d/\hat{v}$, respectively. The corresponding log-relative travel-time residual is therefore $\eta = \log(\hat{T}) - \log(T)$. For a fixed link length, this is equivalently written as $\eta = \log(v) - \log(\hat{v})$,

Table 2
One-hour-ahead prediction error on benchmark datasets.

Dataset	Model	MAE	RMSE
METR-LA	DCRNN	3.590	7.607
	MTGNN	3.490	7.230
Q-Traffic	DCRNN	3.932	6.030
	MTGNN	2.973	4.742

Table 3
Distribution of realized-to-nominal travel-time ratios in the Mandl experiments.

cv	Mean(R)	Std(R)	P95(R)
0.00	1.000	0.000	1.000
0.05	0.988	0.047	1.076
0.25	0.961	0.228	1.436
0.50	0.973	0.456	2.001
1.00	1.090	0.949	3.396

since the length term cancels out algebraically. This equivalence is particularly useful for **METR-LA**, where reliable segment lengths are not directly available for all sensor pairs. In this case, we compute the travel-time log-relative residual directly from the observed and predicted speeds. For **Q-Traffic**, where segment lengths are explicitly provided, the same residual can be obtained either after converting speeds into travel times or directly through the equivalent speed-based expression. The resulting residual pool is then used to generate noisy travel-time forecasts as described in App. E.3.

E.3. zgEmpirical Log-Relative Residual Sampling for Travel-Time Prediction

We pool all empirical prediction residuals on the test sets, across links and time steps, to construct a representative empirical error pool. Since the downstream experiments transfer prediction errors across links and networks with different travel-time scales, we sample log-relative residuals rather than absolute residuals. Specifically, for each realized travel time T and its prediction $\hat{T}$, we define $\eta = \log(\hat{T}) - \log(T) = \log\left(\hat{T}/T\right)$. This log-ratio formulation is appropriate for heteroscedastic prediction errors whose magnitude scales with the level of the predicted variable (Tofallis, 2015); in our setting, this is consistent with traffic scenarios in which the coefficient of variation changes the scale of realized travel times. Finally, independent samples η_{ij}^{ω} are drawn with replacement from this empirical pool for each link (i, j) and scenario ω. Given the realized travel time t_{ij}^{ω} generated under the prescribed traffic-variability level, the corresponding additive travel-time error is defined as $\epsilon_{ij}^{\omega} = t_{ij}^{\omega}\left(\exp\left(\alpha\eta_{ij}^{\omega}\right) - 1\right)$. The predicted link travel times $\hat{t}_{ij}^{\omega}$, used to make network-design decisions, are then computed by adding ϵ_{ij}^{ω} to the actual travel time t_{ij}^{ω}, as $\hat{t}_{ij}^{\omega} = t_{ij}^{\omega} + \epsilon_{ij}^{\omega}$. We do this in both Mandl and Beijing scenarios.

Moreover, because errors are sampled independently for each link and scenario, this first-order model abstracts away the spatial and temporal correlation of real forecast errors, such as corridor-level or incident-level common shocks. The reported results should therefore be read as an evaluation under sampled independent empirical residual scenarios, and correlation-aware forecast-error modeling remains an important extension.

$$\hat{t}_{ij}^{\omega} = t_{ij}^{\omega} + \epsilon_{ij}^{\omega},\ \epsilon_{ij}^{\omega} = t_{ij}^{\omega}\left(\exp\left(\alpha\eta_{ij}^{\omega}\right) - 1\right) \tag{13}$$

Diagnostic check on traffic variability. For the Mandl experiments, we further report the realized-to-nominal travel-time ratio $R_{ij}^{\omega} = t_{ij}^{\omega}/t_{ij}^{0}$, where t_{ij}^{0} is the nominal link travel time and t_{ij}^{ω} is the realized travel time under scenario ω. Table 3 summarizes the distribution of R under different traffic-variability levels. The results show that increasing *cv* mainly amplifies the dispersion and upper tail of realized travel times. Small-*cv* scenarios remain close to nominal conditions, whereas large-*cv* scenarios generate severe congested realizations, as reflected by the increasing standard deviation and 95th percentile of R. In these diagnostics, the prediction-error scale is fixed at $\alpha = 1.0$, corresponding to the empirical error magnitude extracted from the trained prediction model.

F. Algorithmic Framework and Evaluation

F.1. Parameter values for the proposed algorithm

When computing adaptive network $\mathcal{G}_{\text{PT}}$ using the NSGA-III solution method described in the conference paper, we set the algorithmic parameters introduced therein as follows. The minimum path-overlap threshold is set to $J_{\min} = 0.5$, the crossover and mutation probabilities to $P_c = 0.85$ and $P_m = 0.35$, and the number of reference points to $N_{\text{ref}} = 5^3 = 125$, respectively.

F.2. Performance Metrics for Multi-Objective Optimization

We evaluate algorithm performance using three standard metrics for multi-objective optimization: Hypervolume (HV), Inverted Generational Distance (IGD), and Generational Distance (GD). The HV metric quantifies the volume of the objective space dominated by the Pareto solutions:

$$\text{HV}(X, P) = \Lambda\left(\bigcup_{x \in X} v(x, P)\right) \tag{14}$$

where X is the approximation set, P is the reference point, and $v(x, P)$ is the hypervolume dominated by solution x relative to P. Higher HV values indicate better coverage of the objective space. To ensure a fair comparison across algorithms, we report the HV metric calculated relative to a unified reference point. This reference point is constructed by collecting all solutions generated by all algorithms for each objective and selecting the worst (largest) value observed for each objective. The IGD metric measures the closeness of the approximation set to the true Pareto front:

$$\text{IGD}(X, \mathcal{P}^*) = \frac{\sum_{x^* \in \mathcal{P}^*} d(x^*, X)}{|\mathcal{P}^*|} \tag{15}$$

where $\mathcal{P}^*$ is a set of reference points on the true Pareto front, $d(x^*, X)$ is the minimum distance from x^* to any point in X, and $|\mathcal{P}^*|$ is the cardinality of $\mathcal{P}^*$. Lower IGD values indicate better convergence to the true Pareto front. The GD metric complements IGD by measuring how far the obtained solutions are from the true Pareto front:

$$\text{GD}(X, \mathcal{P}^*) = \frac{\sum_{x \in X} d(x, \mathcal{P}^*)}{|X|} \tag{16}$$

where $d(x, \mathcal{P}^*)$ is the minimum distance from a solution x to the true Pareto front $\mathcal{P}^*$, and $|X|$ is the cardinality of the approximation set. Lower GD values indicate better convergence of the approximation set to the true Pareto front. zgAs the true Pareto front $\mathcal{P}^*$ is unknown for the instances considered, we approximate it, following standard practice in multi-objective optimization, by the non-dominated subset of the union of all solutions produced by all compared algorithms across all runs at a given variability level. The HV reference point is constructed analogously from the worst objective values observed across all algorithms (as described above), so that HV, IGD, and GD are all evaluated against a common, algorithm-agnostic reference.

F.3. Implementation Details of the NSGA-III Environmental Selection

The algorithm adopts the reference-point-based environmental selection mechanism of NSGA-III (Deb and Jain, 2014) to maintain diversity across the three objectives in the conference paper. Since all objectives are formulated as minimization problems, normalization is performed on the combined population $R_t = P_t \cup Q_t$. For each objective $j \in \{1, 2, 3\}$, objective-wise lower and upper bounds are computed as $z_j^{\min} = \min_{x \in R_t} Z_j(x)$ and $z_j^{\max} = \max_{x \in R_t} Z_j(x)$, and each objective is linearly scaled to $[0, 1]$ accordingly.

A set of uniformly distributed reference points on the unit simplex is generated following the simplex-lattice design in Deb and Jain (2014). As in standard multiobjective optimization, we say that solution x dominates y if x is no worse than y in all objectives and strictly better in at least one. We employ non-dominated sorting, which partitions the **combined population** $R_t = P_t \cup Q_t$ (comprising both parent and offspring individuals) into a sequence of Pareto fronts $F = \{F_1, F_2, \ldots, F_l\}$. Here, F_1 contains the globally non-dominated solutions F_2 contains solutions dominated exclusively by members of F_1, and F_l represents the *critical front* where the cumulative count of solutions first meets or exceeds the population capacity *PopSize*. Each solution is associated with its nearest reference point by computing perpendicular distances from the solution to the reference lines in the normalized space. The environmental selection process adds complete Pareto fronts F_1 through F_{l-1} sequentially; for the last front F_l, slots are filled by prioritizing solutions associated with reference points that have fewer assigned solutions to ensure diversity.

Algorithm 1 Feasibility-Oriented Path Completion for Line Repair

Input: Substrate graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, endpoints (u, v), visited node set $\mathcal{V}^{\ell}_{\text{used}}$
Output: Connecting path $\mathcal{P}_{u \to v}$ or $\emptyset$

$visited \leftarrow \{u\} \cup \mathcal{V}^{\ell}_{\text{used}}$, $queue \leftarrow \langle u \rangle$, $parent(\cdot) \leftarrow \emptyset$
while *queue* is not empty **do**
  $i \leftarrow queue$.pop_front()
  if $i = v$ **then**
    break
  end if
  for each neighbor j of i in $\mathcal{G}$ **do**
    if $j \notin visited$ **then**
      $visited \leftarrow visited \cup \{j\}$
      $parent(j) \leftarrow i$
      $queue$.push_back(j)
    end if
  end for
end while
if $v \notin visited$ **then**
  return $\emptyset$
end if
$\mathcal{P}_{u \to v} \leftarrow$ Backtrack from v to u using $parent(\cdot)$
return $\mathcal{P}_{u \to v}$

Performance Metric	NSGA-II	NSGA-III	NSVND-II	NSVND-III
Inverted Generational Distance (IGD, lower is better)	347.64	**22.63**	38.90	34.80
IGD Change[a]	–	**-325.01**	-308.74	-312.84
Generational Distance (GD, lower is better)	75.36	**17.17**	24.40	22.80
GD Change[a]	–	**-58.19**	-50.96	-52.56
Hypervolume (HV, higher is better) [b]	30.52×10^9	31.49×10^9	31.23×10^9	31.31×10^9
HV Change[a]	–	**0.97**$\times 10^9$	0.71$\times 10^9$	0.79$\times 10^9$

[a] Improvement/Change are computed relative to NSGA-II (baseline).
[b] HV uses a single fixed reference point shared by all algorithms.

Table 4
Multi-objective algorithm comparison under noise-free travel times ($cv = 0$).

F.4. Algorithm Performance Evaluation

Algorithmic performance is evaluated via the Hypervolume (HV), the Inverted Generational Distance (IGD), and the Generational Distance (GD), which are standard performance metrics in multi-objective evolutionary optimization as surveyed in (Riquelme, Von Lücken and Baran, 2015).[2] The Hypervolume (HV) evaluates both convergence and diversity by measuring the dominated volume in the objective space, where higher values indicate better performance (↑). In contrast, GD and IGD are distance-based indicators that assess convergence to a reference Pareto front, with lower values indicating better convergence (↓).

To evaluate the effectiveness of NSGA-III for zg~~real-time~~forecast-triggered PT redesign, we compare it with NSGA-II (Deb, Pratap, Agarwal and Meyarivan, 2002) and hybrid multi-objective algorithms that combine non-dominated sorting with variable neighborhood descent, based on Mandl's Swiss network. Specifically, we consider two Non-dominated Sorting Variable Neighborhood Descent (NSVND) variants inspired by the integration of NSGA-II with variable neighborhood search strategies (Hansen and Mladenović, 2001). The first variant (NSVND-II) follows the NSGA-II framework while replacing crossover and mutation with variable neighborhood descent, whereas the second variant (NSVND-III) further incorporates an adaptive neighborhood selection mechanism to enhance exploration of the Pareto frontier.

[2] Formal definitions and computational details are provided in App. F.2.

Table 4 presents an algorithm performance comparison for adaptive network optimization assuming no travel time deviation from the nominal ones (cv=0). Although no adaptive redesign is required when $cv = 0$, this setting provides a noise–free benchmark for comparing optimization algorithms. NSGA-III maintains an optimal balance between solution diversity (high hypervolume) and convergence precision.

G. Additional network performance results

zgThe following tables retain the detailed results from the original manuscript that are condensed in the conference paper. Table 5 reports all five tested Mandl line counts; Table 6 reports the Beijing comparison at all tested skip thresholds and variability levels. The latter also reports disconnected demand; travel-time results for stop-skipping concern the reachable OD pairs.

No. of lines	Metric	CV=0	CV=0.05	CV=0.25	CV=0.50	CV=1.00
4	20% improv. share (IS) (%)	5.79 ± 2.99	8.25 ± 4.65	15.40 ± 5.66	24.63 ± 7.91	30.57 ± 11.60
	Op. Cost Red. (%)	10.20 ± 5.31	13.28 ± 7.37	16.86 ± 6.74	24.06 ± 12.01	39.20 ± 12.94
	Line Overlap (%)	84.34 ± 5.44	83.79 ± 7.73	78.83 ± 6.65	77.53 ± 5.27	77.26 ± 5.19
6	20% improv. share (IS) (%)	8.96 ± 4.47	9.53 ± 3.60	16.31 ± 5.10	23.46 ± 7.84	31.50 ± 10.46
	Op. Cost Red. (%)	25.94 ± 4.71	28.94 ± 5.40	32.10 ± 5.69	36.75 ± 9.55	48.90 ± 10.19
	Line Overlap (%)	79.58 ± 2.29	81.20 ± 2.80	78.16 ± 3.94	79.63 ± 4.47	79.16 ± 4.08
8	20% improv. share (IS) (%)	11.52 ± 2.67	13.99 ± 3.90	18.62 ± 5.64	26.84 ± 8.76	35.71 ± 11.53
	Op. Cost Red. (%)	26.97 ± 4.45	29.92 ± 3.84	31.51 ± 5.27	35.35 ± 7.98	45.35 ± 9.23
	Line Overlap (%)	83.58 ± 4.07	84.95 ± 3.07	82.51 ± 4.03	80.80 ± 5.25	81.14 ± 4.23
10	20% improv. share (IS) (%)	8.83 ± 3.15	9.74 ± 4.54	14.27 ± 5.61	21.46 ± 8.88	31.21 ± 11.56
	Op. Cost Red. (%)	33.62 ± 4.18	34.65 ± 4.93	37.23 ± 5.53	40.62 ± 8.71	52.59 ± 9.17
	Line Overlap (%)	77.57 ± 3.40	78.14 ± 2.76	76.96 ± 3.22	78.77 ± 4.09	78.34 ± 3.46
12	20% improv. share (IS) (%)	9.76 ± 3.57	10.34 ± 3.19	14.74 ± 3.51	21.44 ± 6.96	32.44 ± 11.91
	Op. Cost Red. (%)	35.94 ± 4.95	37.51 ± 5.54	39.56 ± 5.18	44.20 ± 6.90	54.27 ± 9.00
	Line Overlap (%)	80.69 ± 2.98	80.95 ± 3.40	80.61 ± 2.78	81.46 ± 3.27	80.77 ± 4.11

No. of lines: number of bus lines; CV: coefficient of variation. 20% improvement share ($IS^{20\%}$): fraction of OD pairs whose travel- time improvement is at least 20%; Op. Cost Red. (OCR): Operational Cost Reduction (reduction in total vehicle-time proxy described in the conference paper). The static route-frequency solutions for each number of bus line are taken from Ahern et al. (2022); entries report mean ± standard deviation over 30 realizations.

Table 5
Performance comparison of the solutions from Ahern et al. (2022) under different variability levels.

Method	Thr. Skip threshold	CV Coeff. var.	LO Line Overlap (%)	OCR Op. cost red. (%)	Avg.TTI Travel-time impr. (%)	Unreach. OD Unreachable OD (%)	Unserv. Dem. Unserved demand (%)
Stop-skip.	25%	0.00	91.16	1.24	3.26	4.89	3.89
		0.05	91.71	1.15	2.73	4.29	3.46
		0.25	56.02	6.58	12.43	29.12	27.39
		0.50	47.58	6.95	14.08	35.28	32.99
		1.00	42.18	12.71	14.71	38.93	36.04
	50%	0.00	95.52	0.61	−0.10	1.71	1.53
		0.05	95.24	0.63	1.94	1.71	1.53
		0.25	80.32	4.59	10.19	11.70	11.40
		0.50	56.13	8.69	19.42	28.63	27.56
		1.00	47.83	12.39	19.71	35.28	32.99
	100%	0.00	98.30	0.32	0.45	0.03	0.02
		0.05	98.30	0.35	0.49	0.03	0.02
		0.25	93.87	1.63	4.08	2.77	2.09
		0.50	78.57	6.68	17.77	12.23	11.61
		1.00	55.88	13.49	15.11	28.05	26.84
Ours	–	0.00	90.08	0.50	11.40	–	–
		0.05	87.34	1.53	11.60	–	–
		0.25	86.51	3.85	16.99	–	–
		0.50	86.39	12.02	18.44	–	–
		1.00	85.42	22.56	25.78	–	–

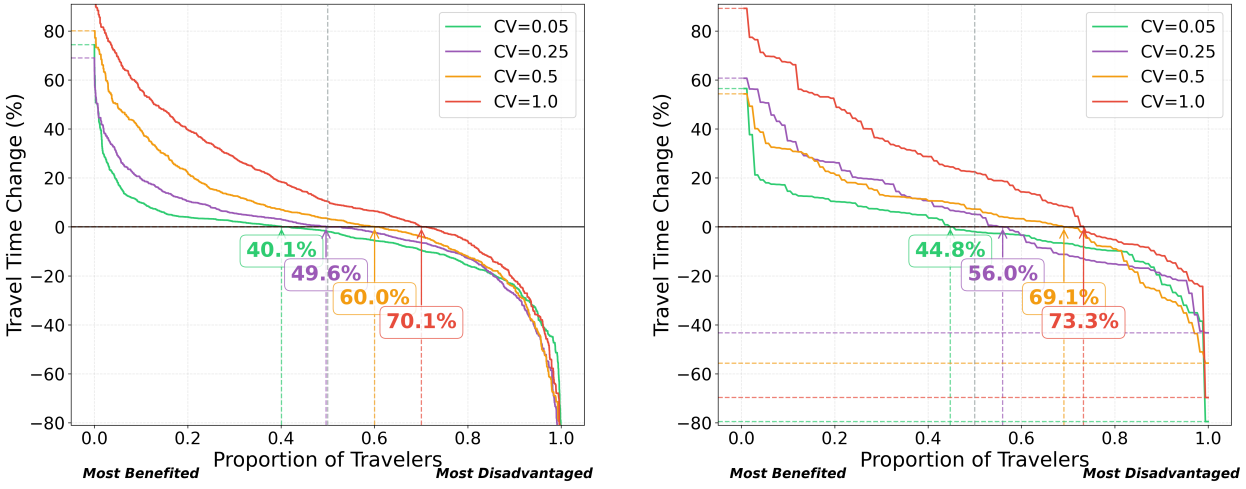
CV: coefficient of variation. Thr.: skip threshold used by the stop-skipping baseline. LO: Line Overlap. OCR: Operational Cost Reduction, measured as the reduction in the vehicle-time proxy. TTI: Travel-Time Improvement, computed as the percentage reduction in the sum of OD travel times relative to the original network. Unreach. OD and Unserv. Dem. report the fraction of OD pairs and demand, respectively, whose connectivity is broken by stop-skipping. These metrics are not applicable to our method, which removes no stops and preserves full OD connectivity.

Table 6

zgComparison of our method and stop-skipping under varying travel-time variability on the Beijing network.

G.1. Trip travel-time distributions

For illustration, in this subsection we report the results for a single randomly chosen realization. zgFigure 2 shows the original manuscript's trip-time distributions for two static Mandl baselines. These panels illustrate one realization, whereas the tables report averages and standard deviations over realizations; the figure should not be interpreted as an aggregate uncertainty estimate.



(a) Comparison with the Ahern et al. static design.

(b) Comparison with the Vermeir et al. static design.

Figure 2: zgTrip travel-time distributions at different variability levels in one randomly chosen realization.

H. Network Performance Metrics

This appendix details the performance indicators used in the experiments.

For each realization $\omega \in \Omega$ of stochastic travel times, we define the following performance metrics: **20% Improvement Share (IS_{20})**, **Operational Cost Reduction (OCR)**, and **Line Overlap (LO)**. Since these metrics vary with the realized scenario ω, they are treated as random variables. In our numerical results, we report their empirical averages (and standard deviations) over 30 independent realizations.

- **20% improv. share (%):** The proportion of OD pairs experiencing a travel time reduction of at least 20% under realization ω.

$$\mathrm{IS}_{20}(\omega) = \frac{\sum_{k\in\mathcal{K}} \mathbb{I}\left(\frac{t_k^0 - t_k^*(\omega)}{t_k^0} \geq 20\%\right)}{|\mathcal{K}|} \times 100\%, \text{ where } \mathbb{I}(\cdot) \text{ is the indicator function.} \tag{17}$$

  Here t_k^0 and $t_k^*(\omega)$ denote the original and adaptive travel times for OD pair k, respectively. Since both depend on random travel-time realizations, IS_{20} is a random variable.

- **Operational Cost Reduction (OCR)**

$$\mathrm{OCR}(\omega) = \frac{\sum_{\ell\in\mathcal{L}} n^\ell t^\ell - \sum_{\ell\in\mathcal{L}} n^{\ell,*}(\omega) t^{\ell,*}(\omega)}{\sum_{\ell\in\mathcal{L}} n^\ell t^\ell} \times 100\%. \tag{18}$$

  Here t^ℓ and $t^{\ell,*}(\omega)$ are the realized travel times of line ℓ in the original and adaptive networks, respectively, and n^ℓ and $n^{\ell,*}$ are the corresponding vehicle allocations. Because t^ℓ and $t^{\ell,*}(\omega)$ vary with the realization, OCR is also a random variable.

- **Line Overlap (LO)** To quantify structural similarity between the original and adaptive networks, we use the Jaccard index:

$$\mathrm{LO}(\omega) = \frac{\left|\mathcal{E}_{\mathrm{PT}} \cap \mathcal{E}_{\mathrm{PT}}^*(\omega)\right|}{\left|\mathcal{E}_{\mathrm{PT}} \cup \mathcal{E}_{\mathrm{PT}}^*(\omega)\right|} \times 100\%. \tag{19}$$

  Here $\mathcal{E}_{\mathrm{PT}}$ and $\mathcal{E}_{\mathrm{PT}}^*(\omega)$ denote the realized edge sets of the original and adaptive networks, respectively. Since these sets depend on the travel-time realization, LO is likewise a random variable.

The numerical results therefore report the *empirical means* ± *standard deviations* of IS_{20}, OCR, and LO over the 30 realizations $\omega \in \Omega$, providing a stochastic assessment of adaptive-network performance.

H.1. Forecast-error Sensitivity Analysis

This appendix provides the formal basis and the full results for the forecast-error sensitivity analysis summarized in the main text. App. H.1.1 defines the two hypervolume-based summaries used to quantify aathe impact of the forecast aa~~value~~error in the optimization output (the *forecast-value headroom* and the *forecast value retained*). App. H.1.2 then applies these metrics to the Mandl instance and reports the detailed results across variability levels and error scales α.

We deliberately grade the entire front rather than any single selected design: the optimizer returns a set of non-dominated trade-offs, and the specific compromise that gets deployed (e.g., the selected compromise solution in the conference paper) is chosen by the operator after the front is produced, according to priorities the algorithm itself does not fix. Tying the sensitivity analysis to one such pre-selected point would conflate two different effects — forecast quality and the arbitrary choice of operating point — since the operator could simply pick a different point on the front to offset a degraded solution, masking a genuine impact of the forecast accuracy on the quality of the solutions (or manufacturing a spurious one). Hypervolume sidesteps this confound by scoring the whole menu of options the operator has to choose from in a single scalar: a drop in HV reflects an actual shrinkage of what is available to the operator, not an artifact of which point happens to be selected.

H.1.1. Forecast-value metrics

We quantify whether the forecast helps the redesign, and by how much. Holding the realized aa~~scenario~~link travel times $\{t_{ij}^{\omega}\}$ fixed, we vary *only* the forecast link times $\hat{t}_{ij}^{\omega}(\alpha)$ handed to the optimizer ($\hat{t}_{ij}^{\omega}(\alpha) = t_{ij}^{\omega} \exp(\alpha \eta_{ij}^{\omega})$) and score the returned Pareto front on the true realized times t_{ij}^{ω}. Since only the error scale α changes, any difference is attributable to forecast quality alone. Each front is graded by its hypervolume HV, the volume it dominates in the space of the three performance metrics (IS_{20}, OCR, LO) defined in Appendix efA.PerformanceMetrics, measured from the single fixed reference point of App. F.2; a larger HV is betteraa ~~and needs no weighting of the three metrics~~. For each seed $s \in \{1, \dots, S\}$ (with $S = 30$ random seeds) and each variability level cv we compute three fronts:

- *No forecast*: aathe optimizer uses the nominal link times t_{ij}^{0}aa, ~~giving the *floor*~~We call the corresponding hypervolume $HV_{s,cv}^{\mathrm{NF}}$ aa~~(lower bound)~~*floor*, since it corresponds to the lower bound.
- *Perfect foresight* ($\alpha = 0$): the optimizer uses the true realized times $t_{ij}^{\omega} = \hat{t}_{ij}^{\omega}(0)$. We denote the corresponding hypervolume by $HV_{s,cv}^{\mathrm{PF}} \equiv HV_{s,cv,0}$ and call it the *ceiling*, since it represents the upper bound.
- *Error-α forecast*: the forecast $\hat{t}_{ij}^{\omega}(\alpha)$ defined above ($\alpha = 1$ the empirical DCRNN level, $\alpha > 1$ a stress test), giving the intermediate hypervolume $HV_{s,cv,\alpha}$.

Because a seed fixes both the realized scenario and the sampled errors η, each $HV_{s,cv,\alpha}$ is a single realization, and all reported 95% confidence intervals reflect variation across the S seeds. We measure the climb from floor to ceiling in two steps.

Step 1: is there anything to gain? The *forecast-value headroom* $H_{s,cv} = \left(HV_{s,cv}^{\mathrm{PF}} - HV_{s,cv}^{\mathrm{NF}}\right)/HV_{s,cv}^{\mathrm{PF}}$ is the floor-to-ceiling gap as a fraction of the ceiling: $H \approx 0$ means even a perfect forecast barely beats nominal times, whereas a large H signals real value to capture. We report the mean $\bar{H}_c$ across seeds with a 95% confidence interval.

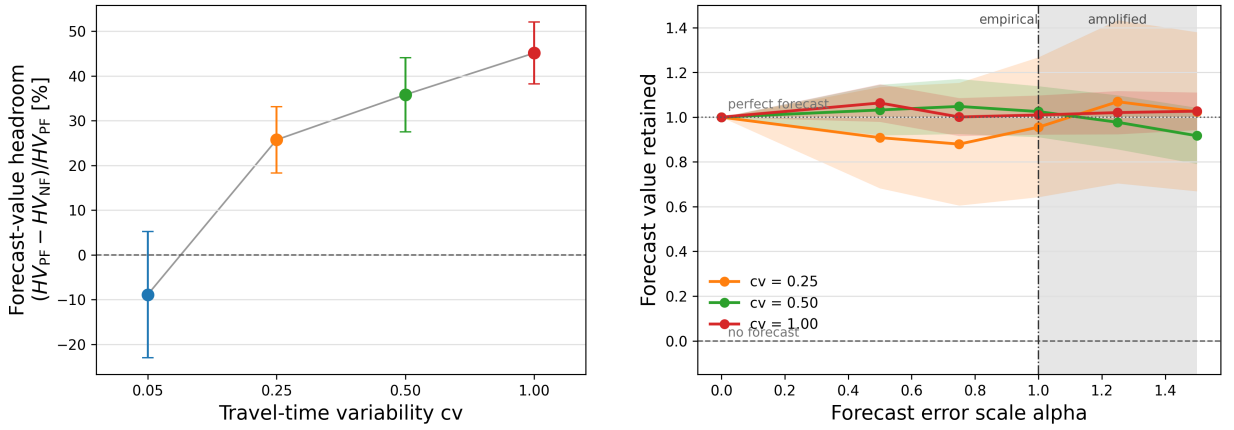
Step 2: how much do we actually capture? The *forecast value retained* $R_{s,cv,\alpha} = \left(HV_{s,cv,\alpha} - HV_{s,cv}^{\mathrm{NF}}\right)/\left(HV_{s,cv}^{\mathrm{PF}} - HV_{s,cv}^{\mathrm{NF}}\right)$ places an error-α forecast on that same scale: $R = 1$ matches perfect foresight, $R = 0$ matches no forecast, and $R < 0$ is worse than no forecast. Since this gap can be tiny for a single seed (making per-seed ratios erratic, unlike $H_{s,cv}$, whose denominator $HV_{s,cv}^{\mathrm{PF}}$ stays bounded away from zero), we aggregate it as a ratio of averages,

$$\bar{R}_{c,\alpha} = \frac{\frac{1}{S}\sum_s \left(HV_{s,cv,\alpha} - HV_{s,cv}^{\mathrm{NF}}\right)}{\frac{1}{S}\sum_s \left(HV_{s,cv}^{\mathrm{PF}} - HV_{s,cv}^{\mathrm{NF}}\right)}, \tag{20}$$

with a 95% confidence interval and $\bar{R}_{c,0} = 1$ by construction. At the lowest variability ($cv = 0.05$) the floor and ceiling are statistically indistinguishable (Fig. 3a), so R is undefined and we omit this level from Fig. 3b.

H.1.2. zg Forecast-value results

Fig. 3 reports the forecast-value metrics of App. H.1.1 for the 4-line Mandl instance. By construction, these metrics separate two otherwise-conflated effects: the redesign opportunity created by travel-time variability (the *headroom*), and the accuracy with which the forecast exploits it (the *retained value*). Fig. 3(a) shows that the headroom is negligible at $cv = 0.05$ and grows with variability, reaching roughly half of the perfect-foresight hypervolume at $cv = 1.0$: a forecast is valuable primarily when variability creates a genuine redesign opportunity, whereas in stable conditions even an oracle adds little because the nominal times t_{ij}^{0} already describe the operating state. Fig. 3(b) shows that the empirical forecast ($\alpha = 1$) retains essentially all of this available value, and moderate amplification ($\alpha = 1.5$) causes no visible collapse, consistent with the intended robustness claim. **The method's gains are therefore driven mainly by the redesign opportunities that variability creates, not by a dependence on high-precision forecasts.**



(a) zgForecast-value headroom $\bar{H}_c$ vs. cv (mean over 30 seeds; 95% CI bars). Headroom is the share of the perfect-foresight hypervolume attributable to using any forecast.

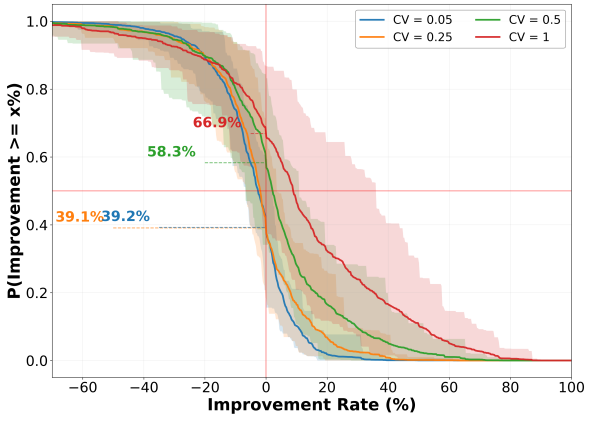
(b) zgForecast value retained $\bar{R}_{c,\alpha}$ vs. error scale α (ratio-of-means; shaded 95% CI). $\alpha = 1$ is the empirical DCRNN error; $\alpha > 1$ is a stress test.

Figure 3: zgHypervolume-based forecast value on the Mandl redesign (mean over 30 seeds): (a) headroom available from any forecast as variability grows; (b) fraction of headroom retained as forecast error is amplified. The method retains essentially all available value at the empirical error level and degrades only under severe amplification.

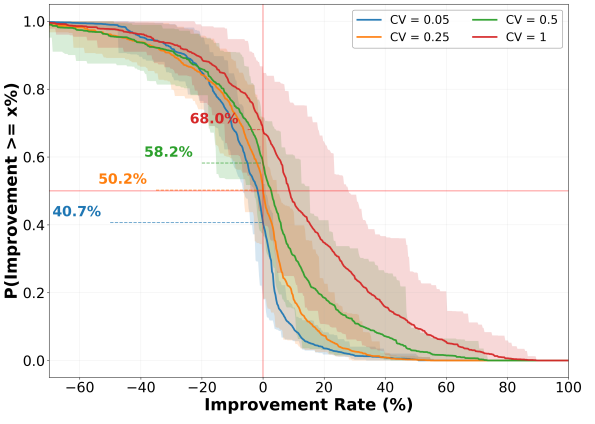
H.2. zgExceedance Probability of OD-Level Improvement Rates by Bus-Line Quantity

aaFor each variability level $cv = 0.05, 0.25, 0.5, 1$ and each random seed $\omega \in \Omega$, we compute the empirical complementary cumulative distribution function (CCDF) of the percentage travel-time improvement over the set of OD pairs. Thus, for a threshold x, the CCDF gives the fraction of OD pairs whose percentage improvement is larger than x. The solid lines in Fig. 4 report the pointwise average of these empirical CCDFs over the 30 seeds. The figure shows that proactive redesign becomes beneficial only under large travel-time variability, namely for ($cv \geq 0.5$). For lower variability levels, it worsens travel times for a larger fraction of OD pairs than it improves.

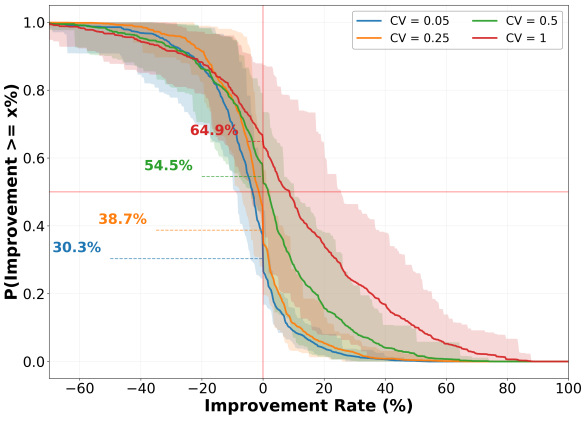
aaFig. 4 ~~reports the trimmed-mean exceedance curves of OD-level improvement rates. For each seed ω, the OD-level improvement rates are first sorted to form a per-seed quantile curve $Q_\omega(p)$, where p denotes the exceedance probability. The extreme seed curves are then removed and the remaining per-seed quantile curves are averaged pointwise. Thus, the plotted line is a trimmed mean of per-seed quantile curves, rather than a curve computed from pooled OD pairs across all seeds. Across line quantities, the curves make it possible to compare both the intensity of improvements and the share of OD pairs receiving a given minimum percentage reduction under different variability levels.~~



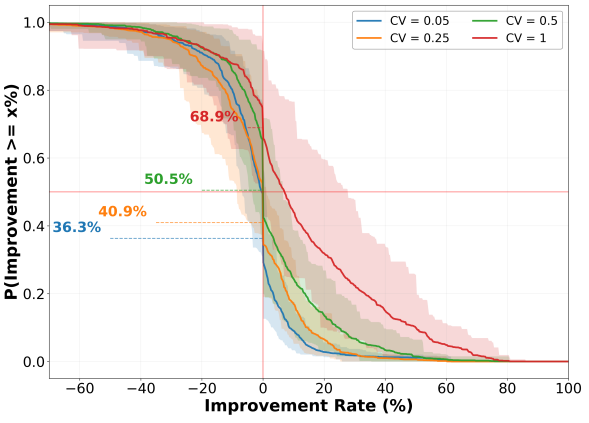
(a) 4 lines



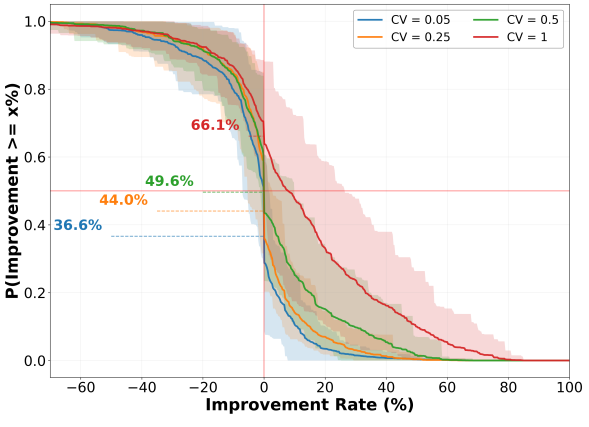
(b) 6 lines



(c) 8 lines



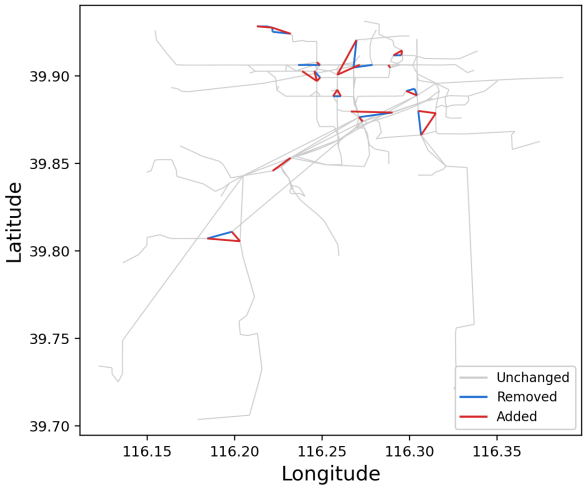
(d) 10 lines



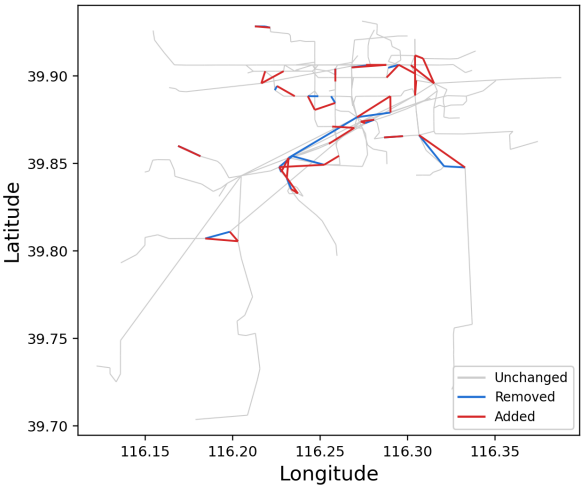
(e) 12 lines

Figure 4: zgTrimmed-mean exceedance probability curves of OD-level improvement rates under different line quantities and variability levels. Each line is computed by first forming one per-seed quantile curve $Q_\omega(p)$ and then averaging the remaining curves after trimming extreme seed curves.

I. Additional Beijing Network Visualisations



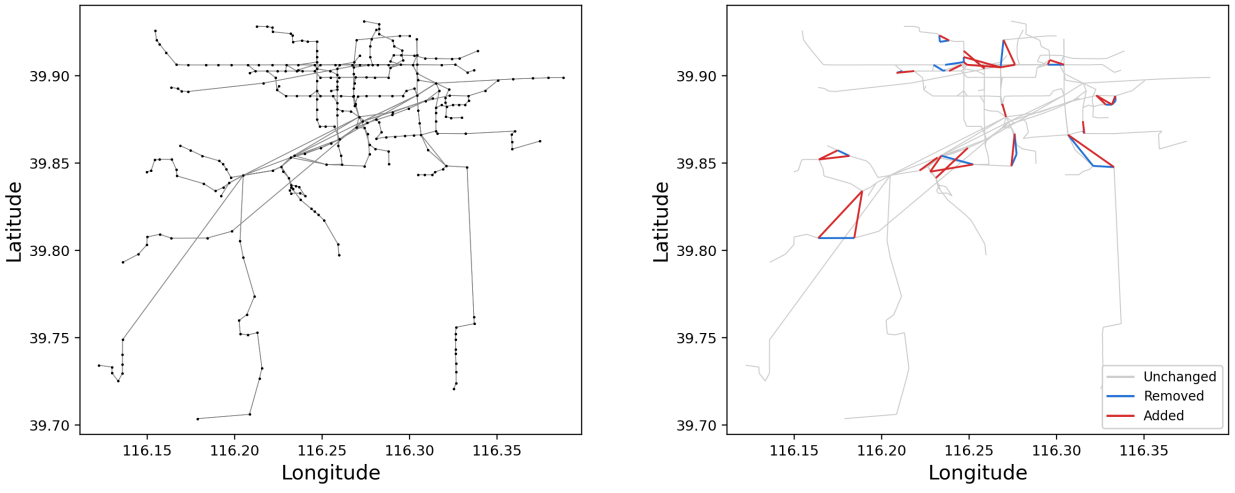
(a) $cv = 0$



(b) $cv = 1$

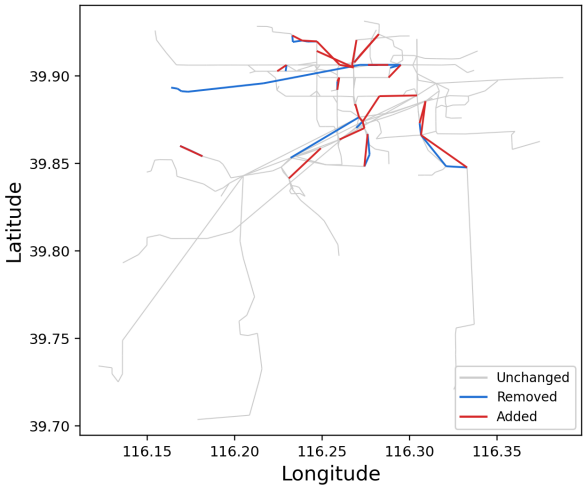
Figure 5: zgAdapted Beijing networks at nominal and high variability. Blue links are removed and red links added relative to the original line assignments, as shown by the plot legends.

zgFig. 6 complements Fig. 5 by reporting the original network and the adapted line layouts at intermediate variability levels. These additional panels show that the redesign remains spatially targeted as cv increases: lines are adjusted around affected corridors while preserving the broad structure of the original service network.

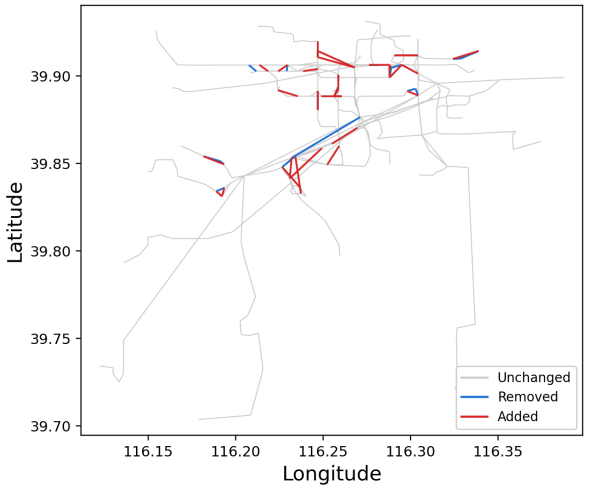


(a) Original network

(b) $cv = 0.05$



(c) $cv = 0.25$



(d) $cv = 0.5$

Figure 6: zgBeijing PT networks: original design and adapted networks under $cv \in \{0.05, 0.25, 0.5\}$.

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